



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Coefficients, P-Values, and Multicollinearity

WEEK 1 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

How sure is each coefficient?

Every coefficient was fitted from 615 bonds. Another 615 would give other numbers.

Two readings that go too far

A small p-value: "this feature matters".

A large p-value: "this feature does nothing".

The recipe behind the spreads

In: rating, years to maturity, an issuer effect, noise.

Not in: sector. Not in: duration.

821 synthetic bonds of 150 invented issuers. Every coefficient today is an association in an invented file.

statsmodels: coefficients plus uncertainty

```
fit = sm.OLS(y_train, sm.add_constant(X_train)).fit()  
fit.params, fit.bse, fit.pvalues
```

```
linear = LinearRegression().fit(X_train, y_train)  
linear.intercept_, linear.coef_
```

12

coefficients

603

residual degrees of freedom

0.8847

R-squared, training rows

Without sm.add_constant: score 0.3055 instead of 0.2016, and no error.

Same least squares coefficients within 1e-9. statsmodels for inference, scikit-learn for pipelines.

Reading the coefficient table

Column	duration	Reads as
coef	0.0197	about 2 percent wider per year, other features fixed
std err	0.0033	typical move from sample to sample
t	5.96	standard errors from zero
$P > t $	0.000	below 0.0005, not zero
interval	0.0132 to 0.0262	values the data are consistent with

What a p-value says

If the true coefficient were zero and the assumptions held: the share of samples giving a t at least this far out.

What it is not

Not the probability the coefficient is real. Not a size. Not a cause. 0.05 is a convention.

Sector: a feature the recipe never used

0 of 9

sector p-values below 0.05

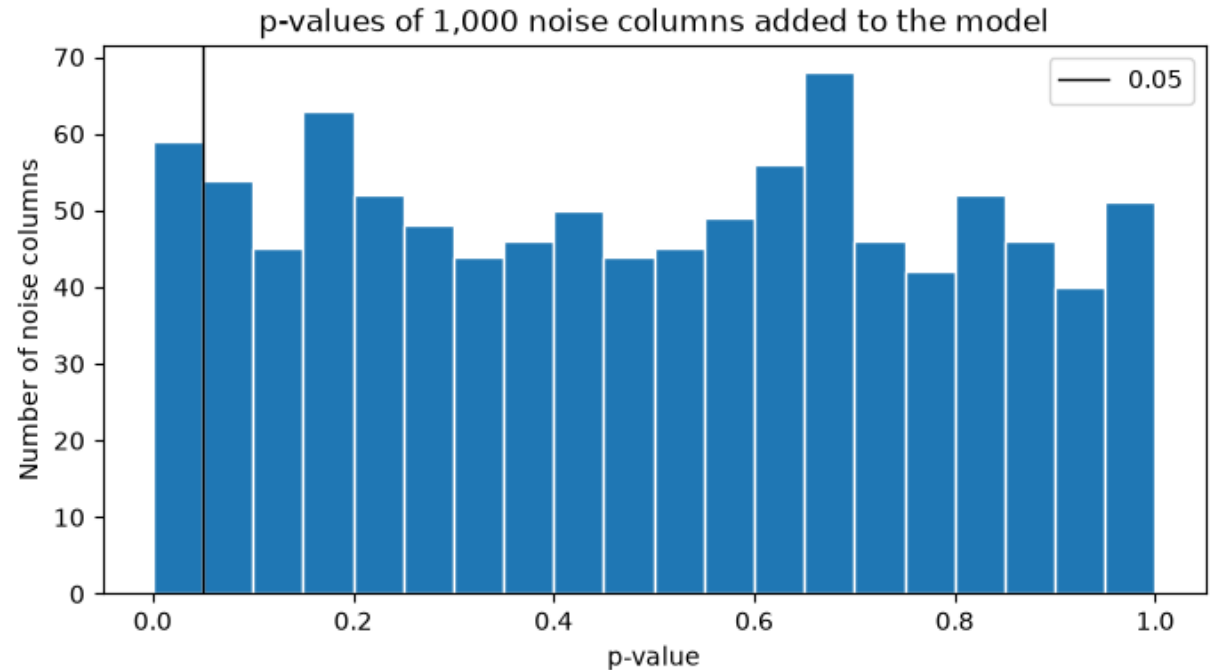
0.283

smallest p (Telecom)

Joint F-test, all nine together

F 0.82, p 0.599

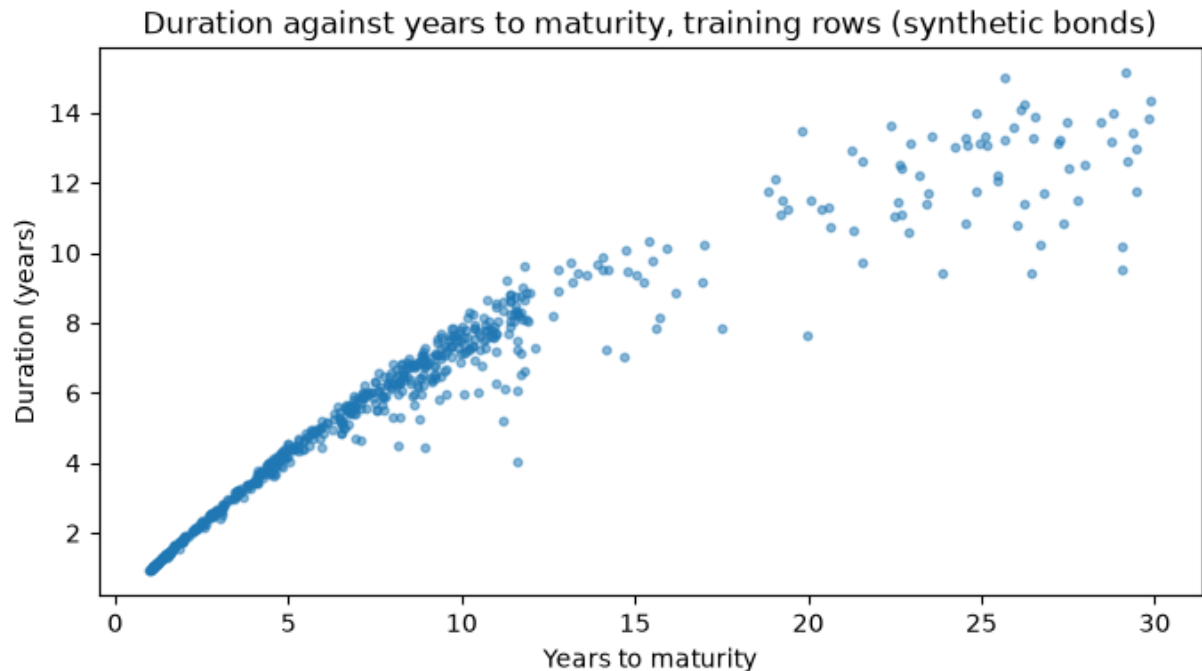
Largest sector p-value: 0.935. A large p does not prove zero.



59 of 1,000 noise columns came in below 0.05 (12 below 0.01): one in twenty, by chance.

Association is not cause

"Holding the other features fixed, a one-unit difference in x goes with a b difference in the fitted y , in this data."



0.9457

correlation of duration and years to maturity

Fair: bonds with longer duration have wider spreads in this file.

Not supported: duration widens the spread. The recipe uses maturity, not duration.

Multicollinearity: duration against years to maturity

Model	Feature	coef	std err	p
Duration only	duration	0.0197	0.0033	< 0.001
Both	duration	0.0067	0.0111	0.545
Both	years_to_maturity	0.0064	0.0052	0.22

Both, tested together

F 18.50

p < 0.001

$VIF_j = 1 / (1 - R^2_j)$ R^2_j : feature j regressed on the others

1.03 to 11.56

duration's VIF, years added

11.35

years to maturity's VIF

3.35x

duration's standard error

Above 5 high, above 10 very high: conventions, not laws.

Perfect collinearity: spread_duration equals duration

spread_duration equals duration on 821 of 821 bonds: the VIF is 1 / 0.

statsmodels: warns, then splits in half

duration 0.0100 + spread_duration 0.0100 = 0.0201, the coefficient of duration alone.

SingularMatrixWarning: the parameters are not uniquely determined.

Excel LINEST: sets one aside (tested)

duration: coefficient 0, standard error 0. spread_duration: 0.0201.

Residual degrees of freedom 612, as if the copy were not there.

Different software, a different arbitrary answer, the same fitted values. Drop the copy.

Dropping a feature by VIF

Highest VIF on the training rows: duration (11.56) against years_to_maturity (11.35). Dropped: the recipe's own column is kept, by luck.

Model	Test R-squared (log)	Test MAE (bp)	Test RMSE (bp)
duration	0.8690	35.9	85.6
duration and years	0.8690	35.2	84.0
duration dropped	0.8684	34.9	83.3

After the drop: highest VIF 2.13; years to maturity 0.0094 per year, standard error 0.0016.

Multicollinearity blurs the reading of coefficients, not the fitted values. The VIF cannot say which column is right.

Bonds of one issuer are not independent

46.0%

of the residual sum of squares shared within an issuer

24.2%

shuffled labels, on average (32.0% at most)

What that does to the table

OLS counts 615 independent bonds; for rating and sector there are about 149 issuers. Standard errors and p-values come out too small; the coefficients are not biased.

The standard fix, cluster-robust standard errors, is a later topic. Day 5 splits the rows by issuer.

Excel side by side

Excel side by side (tested in Excel)

=LINEST(B2:B616,C2:M616,TRUE,TRUE) (Q2)

=T.DIST.2T(ABS(Q2/Q3),\$R\$5) (Q9)

=Q2-T.INV.2T(0.05,\$R\$5)*Q3

=1/(1-INDEX(LINEST(D2:D616,HSTACK(C2:C616,E2:M616),...),3,1))

=((SSR without sectors - \$R\$6)/9)/(\$R\$6/\$R\$5)

=F.DIST.RT(Q19,9,\$R\$5)

=LINEST(B2:B616,HSTACK(C2:D616,O2:O616),TRUE,TRUE)

5 rows by 12 columns, reversed

duration p < 0.001

duration from 0.0132

VIF 1.03

F 0.82

p 0.599

duration: 0, se 0

LINEST returns the columns in reverse: the last X column first, the intercept last.

Save the day: read what a commit changed

3. Experiment log: three log_oas models, with duration, with years to maturity added, and duration dropped

2. Add vif_table, with four tests: VIF, perfect collinearity, OLS against scikit-learn and Excel

1. Start of day 3: mlkit as it stands after day 2

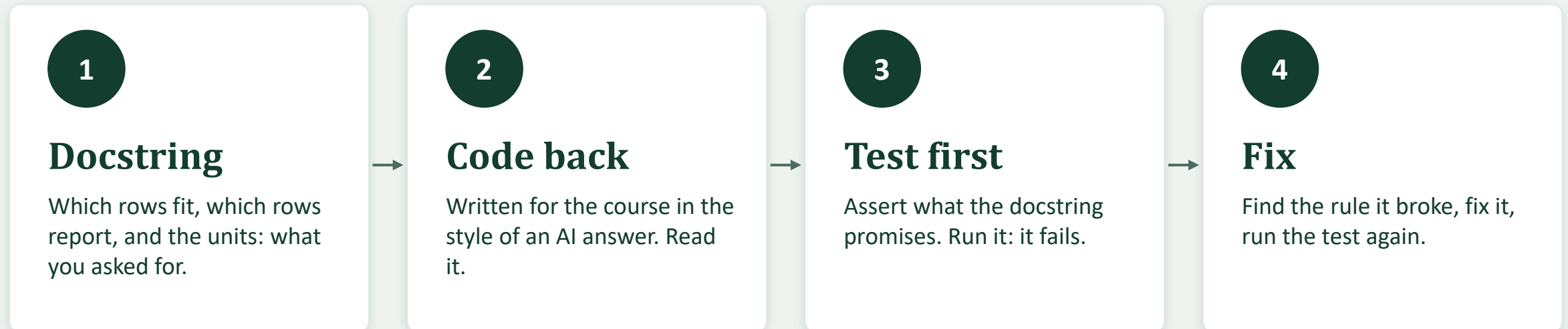
```
git -C "{work}" log -p -1 -- mlkit.py
```

-p: the changes line by line. -1: one commit. -- mlkit.py: only commits that touched the module, so the experiment log is skipped.

14 passed at the end of the day

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: dts (spread duration x OAS) in the features: test R-squared 0.7193 against 0.5529, MAE 38.6 against 57.5 bp.

Code that runs is not the same as code that is right.

What you can do now

- statsmodels: scikit-learn's coefficients plus their uncertainty
- A p-value is not the chance the feature matters
- A large p-value does not prove zero
- Overlap blurs coefficients; the VIF measures it
- Bonds of one issuer: p-values too small

Week 1, Day 4 Regression assumptions and log spreads: the checks behind every p-value